MIL-A-55379A(ER)

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SUPERSEDING
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MILITARY SPECIFICATION

DIRECTION FINDER ANTENNA GROUP

P/O DIRECTION FINDER SET AN/TRQ-30( )(V)

This specification is approved for use by the U.S. Army Communications and Electronics Materiel Readiness Command (USACERCOM), also U.S. Army Electronics Research and Development Command (USAERADCOM), Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers the requirements for the construction and testing of the following units of the Direction Finder Antenna Group which is part of the Direction Finder Set AN/TRQ-30( )(V) herein called "Antenna Group Items". These Antenna Group Items are required to operate in the frequency range of 0.515 to 157.5 MHz with the Receiving Set Radio AN/TRQ-30( )(V) (see 6.3):

Pedestal, Antenna AB-1110( )/TRQ-30(V)
Antenna AS-1523( )/TRQ-30(V)
Antenna AS-1526( )/TRQ-30(V)
Antenna AS-1527( )/TRQ-30(V)
Antenna AS-1528( )/TRQ-30(V)

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, USAERADCOM, Fort Monmouth, N.J. 07703, ATTN: DELET-DP, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5820

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1.1 (cont'd)

Case, Antenna CY-7331( )/TRQ-30(V)
Case, Antenna CY-7332( )/TRQ-30(V)
Case, Antenna Pedestal CY-7333( )/TRQ-30(V)
Compass, Magnetic, Pilot's Standby MS-17983-1
Bracket Assembly, Compass
Cable Assembly, Radio Frequency CG-409H/U(18 ft)

2. APPLICABLE DOCUMENTS.

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids or requests for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	Plastic Sheet and Strip, Thin Gauge, Polyolefin
T-T-871	Twine, Cotton, Wrapping
NN-P-71	Pallets, Materials Handling, Wood (General Construction Requirements)
QQ-S-781	Steel Strapping, Flat
CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied Yarns, Army and Numbered
DDD-T-101	Tape, Linen Finish, Red
NNN-P-40	Paper, Lens
PPP-B-585	Box, Wood, Wirewound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple Walled
PPP-C-843	Cushioning Material, Cellulosic
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Rock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic (and connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain for Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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2.1 Government documents (cont'd)

MILITARY

MIL-P-116	Preservation, Packaging Methods of
MIL-S-1698	Sling, Carrying, Bag and Case
MIL-C-5501	Cap and Plug, Protective Dust and Moisture Seal
MIL-C-5604	Compass, Magnetic, Pilot's Standby
MIL-C-11268	Parts, Materials and Processes Used In Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-19644	Plastic Molding Material (Polystryrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, For Packaging
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification for. Also Include Supplement
MIL-B-43014	Boxes, Water Resistant Paperboard Folding, Set-Up and Metal-Stayed
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging Of

STANDARDS

FEDERAL

FED-STD-751	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads for 40" x 48" Pallets
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1472	Human Engineering Design Criteria for Military Equipment

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2.1 Government documents. (Cont'd)

DRAWINGS

ERADCOM

DLSM-B-879250	Antenna Group p/o Direction Finder Set AN/TRQ-30( )(V)4
DLSM-B-917383	Antenna Pedestal Case CY-7333( )/TRQ-30(V)
SM-A-917475	Acceptance Test Procedure for Antennas AS-1523( ), AS-1526( ), AS-1527( ) and AS-1528( )/TRQ-30(V)

(Copies of specifications, standards, and drawings required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS.

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The Antenna Group Items shall be constructed in accordance with the requirements of this specification and the drawings listed in the USAERADCOM Data Lists DLSM-B-879250 and DLSM-B-917383.

3.3 Parts, materials and process; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply (see 4.4),

3.3.1 Fabric and thread.

3.3.1.1 General. The color of fabric shall be Olive-Drab No. 7. This requirement shall be met immediately after treatment. The color of thread shall match shade S-1 of the United States Army Standard Color Card for Official Standardization Shades of Sewing Thread. Fabric and thread shall be preshrunk, or allowance shall be made for shrinkage in order to provide for satisfactory fit of finished items both before and after they are soaked in tap water at normal room temperature for 24 hours and then dried thoroughly at normal room temperature. Specification and treatment of materials shall be in accordance with MIL-P-11268 paragraph 3.16, Fiberous Material, Organic (see 4.7.10 and 6.4).

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3.3.1.2 Duck. Cotton duck shall be in accordance with Federal Specification CCC-C-419, Type 1, Number 6 duck.

3.3.1.3 Stitches, seams and stitching. Stitches, seams and stitching shall conform to Federal Standard FED-STD-751.

3.3.2 Finish. The Antenna Group Items shall be finished in accordance with MIL-F-14072 and the drawings listed on DLSM-B-879250 and DLSM-B-917383 (see 4.4 and 6.5).

3.4 Marking.

3.4.1 General. Marking shall conform to MIL-M-13231 and MIL-STD-130. Front marking shall be Group I as described in MIL-M-13231 (see 4.4).

3.4.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum dissembly of the equipment.

3.4.3 Serial numbers. The following Antenna Group Items shall have serial numbers:

Pedestal, Antenna AB-1110( )/TRQ-30(V)
Antenna AS-1523( )/TRQ-30(V)
Antenna AS-1526( )/TRQ-30(V)
Antenna AS-1527( )/TRQ-30(V)
Antenna AS-1528( )/TRQ-30(V)
Compass, Magnetic, Pilot's Standby MS-17983-1

3.5 Performance.

3.5.1 HF antenna loop, AS-1523( )/TRQ-30(V). The AS-1523( )/TRQ-30(V) shall be a shielded seven turn loop antenna fed in an unbalanced manner from a 50 ohm transmission line. The AS-1523( )/TRQ-30(V) shall operate in frequency range of 0.515 to 21.0 megahertz (MHz) in five bands listed below:

Bands	Frequency Range
1	0.515 to 1.20 MHz
2	1.03 to 2.60 MHz
3	2.03 to 5.30 MHz
4	3.83 to 9.40 MHz
5	8.46 to 21.0 MHz

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3.5.1.1 HF antenna loop calibration. Each band of the AS-1523( )/TRQ-30(V) shall be tunable continuously over the frequency range specified in 3.5.1. The frequency scale printed on the AS-1523( )/TRQ-30(V) faceplate for each step shall be calibrated such that the actual indicated frequencies are within ± 5% of the mid frequency from one step to the other over the entire frequency range.

3.5.2 VHF antenna loops, AS-1526, AS-1527 and AS-1528/TRQ-30(V). Each of the three VHF antenna loops AS-1526, AS-1527 and AS-1528/TRQ-30(V) is a shielded single turn structure fed in an unbalanced manner from a 50 ohm transmission line. The three VHF antenna loops shall operate in frequency ranges in bands specified below: (see 4.6.5.1).

Bands	VHF Antenna	Frequency Range
A	AS-1526( )/TRQ-30(V)	19 to 55 MHz
B	AS-1527( )/TRQ-30(V)	50 to 100 MHz
C	AS-1528( )/TRQ-30(V)	95 to 157.5 MHz

3.5.2.1 VHF antenna loops calibration. Each loop of 3.5.2 shall cover as a minimum the frequency ranges specified in 3.5.2 (see 4.6.5.2) and continuously tune over the frequency ranges specified.

3.5.3 Direct-to-reciprocal error. The electrical displacement of the direct null to the reciprocal null shall be 180 degrees ± 5 degrees. This limit shall be met when the antenna(s) is resonated (see 4.6.2).

3.5.4 Null depth. The nulls shall equal or exceed 30 db, down from a point of maximum antenna response (see 4.6.3).

3.5.5 Instrument error. The total instrument error introduced by the loop antennas, compass, azimuth scale and mechanical keying of the mast assemblies shall not exceed ± 2 degrees (see 4.6.4).

3.6 Environmental. The Antenna Group Items shall meet the following service conditions without degradation in specified performance of 3.5 unless specified herein (see 4.7).

3.6.1 Temperature.

3.6.1.1 Low temperature. The Antenna Group Items shall be operable without degradation in specified performance at ambient temperature as low as -40 degrees F and withstand exposure to ambient air temperature as low as -70 degrees F. (see 4.7.1.1).

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3.6.1.2 High temperature. The Antenna Group Items shall be operable without degradation in specified performance at ambient temperatures as high as +131 degrees F and withstand exposure to ambient air temperature as high as +160 degrees (see 4.7.1.2).

3.6.2 Altitude. The Antenna Group Items shall be operable without degradation in specified performance at altitude up to 10,000 feet above sea level and withstand air transportation at altitude up to 17,5000 feet above sea level (see 4.7.2).

3.6.3 Humidity. The Antenna Group Items shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to relative humidities as high as 100 percent at all ambient air temperatures to +85 degrees F and relative humidities corresponding to a dew point of +85 degrees F at all temperatures between +85 degrees F and +120 degrees F, and low relative humidity of 5 percent at +120 degrees F. (see 4.7.3).

3.6.4 Sand and dust. The Antenna Group Items shall withstand, in both operating and nonoperating conditions, exposure to fine sand and dust particles with wind speeds of up to 20 knots. After testing as specified in 4.7.4, the equipment shall meet the specified performance requirements of 3.5.3, 3.5.4 and 3.5.5.

3.6.5 Salt fog. The Antenna Group Items shall withstand the salt fog test of 4.7.5. Upon completion of test, the Antenna Group Items when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surfaces; or exfoliation of metal. There shall be no clogging or binding of controls or other moving parts after subjecting the Antenna Group Items to the 48 hour salt fog exposure period or after subjection to the subsequent 48-hour drying period.

3.6.6 Fungus. After the fungus test of 4.7.6 the Antenna Group Items shall be visually examined, using a 10-power magnifier. The Antenna Group Items shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.6.3) as evidenced by growth colonization (see 6.6.2) which includes branching (see 6.6.1) and sporulation (see 6.6.4) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.6.5) shall not be included in this evaluation.

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3.6.7 Leakage (immersion). The antenna loops and control boxes shall show no evidence of leakage after they are tested as specified in 4.7.7.

3.6.8 Rain. After testing as specified in 4.7.8, the Antenna Group Items shall meet the requirements of 3.5.3, 3.5.4 and 3.5.5.

3.6.9 Vibration, bounce and shock. The Antenna Group Items shall meet the requirements of Table I. (see 4.7.9).

TABLE I

Para. Heading	Test Para	Performance After Test
Shock, bench handling	4.7.9.1	Specified Performance. No physical damage. 1/
Bounce, loose cargo	4.7.9.2	Specified Performance. No physical damage except for minor surface abrasions. 1/ & 2/.
Shock; transit drop	4.7.9.3	Specified Performance. Minor physical damage only. 1/.

1/ After testing, the equipment shall meet specified performance for the following:

Paragraph	
3.5.3	Direct-to-Reciprocal Error
3.5.4	Null Depth
3.5.5	Instrument Error

2/ During resonant search, the amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 hertz. During the resonant search, the antenna loop and the control box shall both be securely clamped to the holding fixture so that they do not move relative to each other. The purpose of this search is to test and observe internal structures and components.

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3.7 Interchangeability. Like units, assemblies, sub-assemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.8.1 and 4.8.2).

3.8 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268

General requirements for plastic material and plastic parts.
Wiring and cabling, Including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Flux and cleaning agent for soldering
Process for soldering
Cleaning
Riveting
General requirements for securing of parts
Cleaning after welding
Mechanical operation of controls
Process for welding
Self-tapping screws

b. In this specification:

3.3.2 Finish

3.9 Systems safety engineering.

3.9.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection. (see 4.9).

3.9.2 Edge rounding exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.05 inch (1.3 mm).

3.9.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.10 Human factors engineering. Design configuration as affecting operation, maintainability and handling shall be maintained through compliance of configuration criteria to the requirements of MIL-STD-1472.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging). (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging.(see 4.10).

4.3 First article. Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish four (4) First Article units of the complete Antenna Group.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in subsidiary documents covering the items listed in 4.4 and the inspections specified for Group A, Group B, and Group C (see Table III, IV and V respectively). The inspection shall be performed in the following order: 4.4, Group A and Group B for all units and designated units for specific Group C tests in the order shown in Table II.

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TABLE II. First article inspection.

Notes - Inspections	Req Para.	Test Para	Items To Be Inspected: Unit 1	Unit 2	Unit 3	Unit 4	Antennas	Pedestal	Cable	Canvas Bags
1/ 1. Inspections Covered by Subsidiary Documents		4.4	Inspections to be performed on all units				X	X	x	x
1/ 2. Group A Inspection		4.5.1	Inspections to be performed on all units				x	x	x	x
1/ 3. Group B Inspection		4.5.2	Inspections to be performed on all units				x	x		
2/ 4. Group C Inspection		4.5.3								
Environmental:										
Low temperature	3.6.1.1	4.7.1.1	1				x	x	x	
High temperature	3.6.1.2	4.7.1.2	2				x	x	x	
Altitudes	3.6.2	4.7.2	3				x	x	x	
Shrinkage (Canvas)	3.3.1.1	4.7.10		1						x
Humidity	3.6.3	4.7.3	4				x	x	x	x
Sand & dust 3/	3.6.4	4.7.4			2		x	x	x	
Salt fog 3/	3.6.5	4.7.5				3	x	x	x	
Fungus 3/	3.6.6	4.7.6		2			x	x	x	x
Leakage (immersion)	3.6.7	4.7.7			1		x		x	x
Rain	3.7.8	4.7.8				2	x	x	x	
Shock, bench handling	3.6.9	4.7.9.1				1	x	x	x	
Bounce, loose cargo	3.6.9	4.7.9.2	5				x	x	x	x
Transit drop	3.6.9	4.7.9.3		3			x	x	x	x

NOTES: 1/ The inspections 1 through 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

2/ The numbers in the unit columns in the table covered by Section 4 (Group C Inspection) specify the order of inspections for the indicated unit.

3/ The equipment shall be thoroughly washed, cleaned, dried and refurbished, if necessary, before proceeding with subsequent inspection.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification.

ITEM	WHERE REQUIRED
Finish, Protective	3.3.2
Marking	3.4
Parts, Materials and Processes	3.3

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the general inspection levels, and AQLs indicated in Table III.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

TABLE III. Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.8	4.9	1.5%	4.0%
Electrical:				
Direct-to-reciprocal error	3.5.3	4.6.2	1.0%	*
Null depth	3.5.4	4.6.3	1.0%	*
Instrument error	3.5.5	4.6.4	1.0%	*
Loop calibration	3.5.1.1			
	3.5.2.1	4.6.5	1.5%	*

*All electrical defects will be considered major.

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4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and to the special inspection level of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans for the AQL's listed in Table IV shall be S-4. All defects shall be considered major.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in any order satisfactory to the Government.

TABLE IV. Group B inspection.

Inspection	Req. Para	Test Para.	AQL
Interchangeability	3.7	4.8	4.0%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. Sample units shall be selected in accordance with 4.5.3.1. The inspection shall consist of the inspection specified in Table V.

4.5.3.1 Sampling for Group C inspection. Four samples of the complete equipment shall be selected at random, 1 sample for each Subgroup I through and including Subgroup IV in Table V. The first samples shall be selected at the start of production from the first quality conformance inspection lot. These samples shall constitute the Group C requirements for the first 25 units produced. Thereafter, 4 samples of the complete equipment shall be selected at random, 1 sample for each Subgroup I through IV in Table V from every 75 units or fraction thereof greater than one-half. If the fraction is less than one-half, the fraction shall be considered part of the previous Group C lot for sampling purposes. Salt fog and fungus shall not be repeated after the first production lot Group C inspection.

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TABLE V. Group C inspection.

Inspection	Req. Para.	Inspection Para.
Subgroup I		
Altitude	3.6.2	4.7.2
Shrinkage	3.3.1.1	4.7.10
Humidity	3.6.3	4.7.3
Subgroup II		
Low temperature	3.6.1.1	4.7.1.1
High temperature	3.6.1.2	4.7.1.2
Vibration	3.6.9	4.7.9
Subgroup III		
Shock, bench handling	3.6.9	4.7.9.1
Bounce, loose cargo	3.6.9	4.7.9.2
Transit	3.6.9	4.7.9.3
Subgroup IV		
Sand and dust 1/	3.6.4	4.7.4
Salt fog 1/	3.6.5	4.7.5
Fungus 1/	3.6.6	4.7.6
Leakage (immersion)	3.6.7	4.7.7
Rain	3.6.8	4.7.8

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order acceptable to the Government.

4.5.3.3 Group C failure. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are re-subjected to and pass Group A and Group B inspection, in that order.

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4.6 Performance.

4.6.1 Standard test conditions. Unless otherwise specified, the electrical tests shall be conducted under the following conditions:

a. All tests shall be conducted under conditions of normal room ambient temperature and humidity as defined in MIL-STD-810.

b. The electrical tests of 4.6.2, Direct-to-Reciprocal-Error, 4.6.3, Null Depth and 4.6.4 Instrument Error shall be conducted at a suitable, interference free field site which is clear of reflectors, re-radiators and absorbers, such as wires, fences, buildings, towers, power lines, etc. However, these tests will be permitted in a screen room or other area for Group A testing only if a correlation is made which is acceptable to the Government.

c. The electrical tests cited in 4.6.1b above shall be performed with an Empire Devices NF-105 Field Strength Meter or equivalent. These tests will be permitted, however, using a receiver having a nominal 50 ohm input impedance and a 1 micro-volt sensitivity at a 10 db signal plus noise to noise ratio, if correlation is made which is acceptable to the Government.

d. Received signals, for test purposes, shall be 30%, 1000 Hz amplitude modulated.

4.6.2 Direct-to-reciprocal error. The direct-to-reciprocal error shall be as specified in paragraph 3.5.3 when tested in accordance with 3.3 of Test Procedure SM-A-917475.

4.6.3 Null depth. The null depth shall be as specified in paragraph 3.5.4 when tested in accordance with 3.4 of Test Procedure SM-A-917475.

4.6.4 Instrument error. The instrument error shall be as specified in paragraph 3.5.5 when tested in accordance with 3.5 of Test Procedure SM-A-917475.

4.6.5 Loop calibration.

4.6.5.1 HF loop calibration. The HF loop calibration shall be as specified in paragraph 3.5.1.1 when tested in accordance with 3.1 of Test Procedure SM-A-917475.

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4.6.5.2 VHF loop calibration. The VHF loop calibration shall be as specified in paragraph 3.5.2.1 when tested in accordance with 3.2 of Test Procedure SM-A-917475.

4.7 Environmental. The Antenna Group Items shall be tested in accordance with the methods and procedures of MIL-STD-810 given below. The limits of allowable performance degradation shall be as specified in 3.5 of this specification, and shall form the criteria for passing or failing each environmental test unless stated otherwise. Successful completion of any particular test shall be considered evidence of meeting the corresponding environmental condition.

4.7.1 Temperature.

4.7.1.1 Low temperature. The Antenna Group Items shall be subjected to the test of Method 502.1, Procedure 1 of MIL-STD-810. The storage temperature shall be -70 degrees F and the operating temperature shall be -5 degrees F. Prior to testing, the Antenna Group Items shall meet full specification performance as specified in 3.5.3, 3.5.4 and 3.5.5.

4.7.1.2 High temperature. The Antenna Group Items shall be subjected to the test of Method 501-1, Procedure II of MIL-STD-810. The operating temperature shall be +131 degrees F. Prior to testing, the Antenna Group Items shall meet full specification performance as specified in 3.5.3, 3.5.4 and 3.5.5.

4.7.2 Altitude. The Antenna Group Items shall be subjected to the test of Method 500.1, Procedure 1 of MIL-STD-810, except Step 2 shall be 17,500 feet. Prior to testing, the Antenna Group Items shall meet full specification performance as specified in 3.5.4, 3.5.4 and 3.5.5.

4.7.3 Humidity. The Antenna Group Items shall be subjected to the test of Method 507.1, Procedure II of MIL-STD-810. In step 6, measurements shall be taken during the first 5 hours of each cycle. Prior to testing, the Antenna Group Items shall meet the full specification performance as specified in 3.5.3, 3.5.4 and 3.5.5 (see 3..3)

4.7.4 Sand and dust. The Antenna Group Items, shall be tested in accordance with Method 510.1, Procedure 1 of MIL-STD-810. Prior to testing, the Antenna Group Items shall meet full specification performance as specified in 3.5.3, 3.5.4 and 3.5.5 (see 3.6.4).

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4.7.5 Salt fog. The Antenna Group Items shall be tested in accordance with Method 509.1, Procedure 1 of MIL-STD-810. Operation of the Antenna Group Items shall not be required, and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the test items surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface (see 3.6.5).

4.7.6 Fungus. The Antenna Group Items shall be subjected to the fungus test specified in Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization (6.6.2) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment.

4.7.7 Leakage (immersion). The Antenna Group Items as prepared for field transport, shall be tested in accordance with Method 512.1, Procedure I of MIL-STD-810. Prior to testing, the Antenna Group Items shall meet full specification performance as specified in 3.5.3, 3.5.4 and 3.5.5. (see 3.6.7).

4.7.8 Rain test. The Antenna Group Items (excluding canvas bags) shall be tested as follows:

a. Dry at 150° ± 5°F for 48 hours.

b. Condition for 4 hours at 77° ± 5°F and 40 to 60% relative humidity.

c. Take pretest data. The Antenna Group Items shall meet full specification performance for the measurements specified in 3.5.3, 3.5.4 and 3.5.5.

d. Test the Antenna Group Items in accordance with Method 506.1, Procedure I, of MIL-STD-810. During each 30 minute rain period, the wind source shall be maintained at the 40 miles per hour volocity for 15 minutes after which the wind source shall be turned off. Complete final measurements within 1 hour after the rain has been discontinued. The Antenna Group Items shall meet the requirements of 3.5.3., 3.5.4 and 3.5.5 and applicable requirements of MIL-STD-810 after test. No measurements are required during this test (see 3.6.8).

4.7.9 Vibration, bounce and shock.

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4.7.9.1 Shock test, bench handling. The antennas, excluding canvas bags, shall be tested in accordance with Method 516.2, Procedure V, of MIL-STD-810. Prior to testing, the Antenna Group Items shall meet full specification performance for the measurements specified in 3.5.3, 3.5.4 and 3.5.5. As a practical matter, since there are only 2 major faces, there will be a total of 8 drops. Pedestal assemblies, with compass attached, will be similarly tested (see 3.6.9).

4.7.9.2 Bounce, loose cargo, including vibration, resonance search. The antennas with mast in the canvas bags shall be tested in accordance with Method 514.2, Procedure XI, Part 2 of MIL-STD-810. Prior to testing, the Antenna Group Items shall meet full specification requirements as specified in Table I (see 3.6.9).

4.7.9.3 Transit drop. The Antenna Group Items in its canvas bags shall be tested in accordance with Method 516.2, Procedure II, of MIL-STD-810. Prior to testing, the Antenna Group Items shall meet full specification performance for the measurements specified in Table I (see 3.6.9).

4.7.10 Shrinkage. Fit the Antenna Group Items into the canvas bags prior to the shrinkage test to determine adequate fit. Soak canvas and fabric items in tap water at room temperature for 24 hours and dry thoroughly at room temperature. Canvas bags shall meet the requirements of 3.3.1.1. before and after soak.

4.8 Interchangeability.

4.8.1 Inspection for dimensional interchangeability. The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, keying pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major units.

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4.8.1 Inspection for dimensional interchangeability (cont'd).

e. Size and form of special threads.

4.8.2 Electrical interchangeability. All like modules and replaceable parts will be physically and electrically interchanged without modification of such items and the performance test of 4.6. However, electrical retuning and realignment of Antenna Group Items during production assembly operations is permissible.

4.9 Visual and mechanical inspection. The Antenna Group Items shall be examined for defects listed in requirements paragraph 3.8 and in MIL-STD-252. An inspection shall also be performed to verify compliance with those portions of 3.9 (Systems Safety Engineering) which can be determined visually.

4.10 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of paragraph 5 herein. The rough handling test will be performed only when specified (see 6.2).

4.11 Human engineering verification. Maintenance of design configuration to the requirements of 3.10 shall be verified by visual and mechanical inspection.

5. PACKAGING.

5.1 Preservation. Preservation shall be Level A, B or Commercial as specified (see 6.2.e).

5.1.1 Level A.

5.1.1.1 Cleaning. Antenna Group Items or Units of shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Antenna Group Items or Units of shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. Not required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be individually unit packed Method IC-1.

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5.1.1.4.2 Ground anchor assemblies. Ground anchor assemblies shall be unit packed Method III as follows: Wind the nylon cord around each individual stake and secure with twine conforming to T-T-871, type II. Bundle three each secured ground anchor assemblies together and secure with twine conforming to T-T-871, type II.

5.1.1.4.3 Carrying strap. Each carrying strap shall be individually unit packed Method III as follows: Wind the strap in a tight coil and secure with twine conforming to T-T-871, type II.

5.1.1.4.4 Pedestal, Antenna AB-1110( )/TRQ-30(V) in Case CY-7333( )/TRQ-30(V). Each antenna pedestal shall be individually unit packed Method III as follows: Place the items, unit packed as specified in 5.1.1.4.2 and 5.1.1.4.3, in the outer pocket of the case. Stow the remaining items of the antenna pedestal in their designated locations within the case and secure the cover. The secured case with contents shall be cushioned on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, designed to protect the unit from shock and vibration hazards during handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned item shall be inclosed within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

5.1.1.4.5 Compass, magnetic. Each magnetic compass shall be individually unit packed Method III as follows: Cushion the item and inclose within a fiberboard box as specified in 5.1.1.4.4.

5.1.1.4.6 Cable Assembly CG-409H/U. Each cable assembly shall be individually unit packed in accordance with MIL-C-55442.

5.1.1.4.7 Antenna AS-1523( )/TRQ-30(V), AS-1526( )/TRQ-30(V), AS-1527( )/TRQ-30(V) and AS-1528( )/TRQ-30(V). Each loop antenna shall be individually unit packed Method III as follows: A protective cap conforming to MIL-C-5501 shall be placed over the cable connector and the antenna cushioned and inclosed within a fiberboard box as specified in 5.1.1.4.4.

5.1.1.4.8 Case, Antenna CY-7331( )/TRQ-30(V) and CY-7332( )/TRQ-30(V). Each antenna case shall be individually unit packed Method III as follows: Enclose case in a close-fitting box conforming to PPP-B-636, W5c, style OPF. Close the box in accordance with the appendix to the box specification.

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5.1.1.4.9 Antenna Group Items. The units of the Antenna Group Items, unit packed as specified in 5.1.1.4.4 through 5.1.1.4.8, shall be consolidated into an individual close fitting fiberboard box conforming to PPP-B-636, W5c. Fill all voids with cushioning material conforming to PPP-C-843, type II. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the centents directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2.e). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Antenna Group Items or Units of, bearing the same stock number and unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

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5.2.2 Level B. A quantity of Antenna Group Items or Units of, bearing the same stock number and unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film conforming to L-P-378, type IV. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of Antenna Group Items or Units of, bearing the same stock number and unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

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6. NOTES.

6.1 Intended use. These antennas are used for a portable, man-pack direction finder set, specifically for use with the AN/TRQ-30( )(V).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Marking and shipping of equipment.

c. Place of final inspection.

d. Technical literature required.

e. Level A, Level B or Commercial preservation and packing (see Section 5).

f. When rough handling test is required (see 4.10).

6.3 Versions of Antenna Groups. The following combination of one (1) each of the items listed in 1.1 comprise the (V)1, (V)2 and (V)4 versions of the Antenna Group AN/TRQ-30( )(V). (It is to be noted that (V)3 version does not exist).

6.3.1 Antenna Group AN/TRQ-30( )(V)1.

Pedestal, Antenna AB-1110( )/TRQ-30(V)
Antenna AS-1523( )/TRQ-30(V)
Compass, Magnetic, Pilot's Standby MS-17983-1
Bracket Assembly, Compass
Cable Assembly, Radio Frequency CG-409H/U
Case, Antenna CY-7331( )/TRQ-30(V)

6.3.2 Antenna Group AN/TRQ-30( )(V)2.

Pedestal, Antenna AB-1110( )/TRQ-30(V)
Antenna AS-1526( )/TRQ-30(V)
Antenna AS-1527( )/TRQ-30(V)
Antenna AS-1528( )/TRQ-30(V)
Compass, Magnetic, Pilot's Standby MS-17983-1
Bracket Assembly, Compass
Cable Assembly, Radio Frequency CG-409H/U
Case, Antenna CY-7332( )/TRQ-30(V)

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6.3.3 Antenna Group AN/TRQ-30( )(V)4.

Pedestal, Antenna AB-1110( )/TRQ-30(V)
Antenna AS-1523( )/TRQ-30(V)
Antenna AS-1526( )/TRQ-30(V)
Antenna AS-1527( )/TRQ-30(V)
Antenna AS-1528( )/TRQ-30(V)
Compass, Magnetic, Pilot's Standby MS-17983-1
Bracket Assembly, Compass
Cable Assembly, Radio Frequency CG-409H/U
Case, Antenna CY-7331( )/TRQ-30(V)
Case, Antenna CY-7332( )/TRQ-30(V)

6.4 Color of textiles. Standard samples of Olive Drab No. 7 may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply Agency, Philadelphia, Pennsylvania. When requesting samples, indication shall be given as to type of material concerned (see 3.3.1.1 and 3.3.1.2).

6.5 Color. The color chip furnished by the procuring agency (see 3.3.2) will be color chip No. 34087, and may be obtained upon request to Commander, U.S. Naval Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

6.6 Definitions.

6.6.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.6.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitues abundant growth.

6.6.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.6.4 Sporulation. Sporulation is the formulation of minute unicellular reproductive or dormant bodies, called spores.

6.6.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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6.6.6 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-A-55379A (ER) DIRECTION FINDER ANTENNA GROUP P/O DIRECTION FINDER SET AN/TRQ-30 ( ) ( )

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